Printed Pages: 01 **Sub Code: BP -813ET**

Paper Id: 236725 **Roll No.** | | | | | | | | | | |

B PHARM
(SEM VIII) THEORY EXAMINATION 2022-23
PHARMACEUTICAL PRODUCT DEVELOPMENT

Time: 3 Hours ***Total Marks: 75***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **10 x 2 = 20**

a. What do you mean by product development?
b. Mention two quality control tests of parenteral dosage form.
c. Name three non aqueous solvents used in pharmaceutical liquids.
d. Mention two uses of Polyethylene Glycol.
e. Differentiate between lubricant and anti-adherents as per their use.
f. Write two examples of enteric coating polymers.
g. What is Fractionate Factorial Design?
h. Write full forms of terms CMA and CPP used in QbD.
i. Name two problems associated with glass as a packaging material.
j. Mention two ideal characteristics of a packaging material.

SECTION B

2. Attempt any *two* parts of the following: **2 x 10 = 20**

a. Elaborate on various excipients used in formulating a semisolid dosage form along with suitable examples.

b. Discuss various material used for the packaging of pharmaceutical products? Explain the criteria of their selection.

c. What do you mean by optimization? Discuss various optimization techniques used in pharmaceutical product development with special reference to factorial designs?

SECTION C

3. Attempt any *five* parts of the following: **7 x 5 = 35**

a. Explain various objectives of Pharmaceutical Product development.

b. Explain various suspending and emulsifying agents with examples.

c. What are the various applications of cyclodextrins in pharmaceutical products?

d. Discuss various excipients used in parenteral products with examples.

e. Define QbD. What are the various steps of QbD applied in product development?

f. Discuss various quality control tests of glass containers as a packaging material.

g. Discuss regulatory requirement for stability assessment of pharmaceutical product.

QP23EP1_311 | 21-06-2023 13:56:28 | 115.243.172.58

QP23EP1_311 | 21-06-2023 13:56:28 | 115.243.172.58